

Edge-Assisted Smart Traffic Signal Regulation via Local Artificial Intelligence and Emergency Vehicle Preemption

Yash*, Sri HAri†

Department of Computer Science and Engineering
BITM ENGINEERING COLLEGE, Ballari, Karnataka, India
Email: *yash1@gmail.com, †ZOOOnIOR@gmail.com

Abstract—Conventional urban traffic management infrastructure predominantly relies on static, fixed-time signal cycles that are fundamentally unequipped to adapt dynamically to real-time spatial and temporal shifts in traffic congestion. While cloud-centralized artificial intelligence (AI) paradigms offer high computational performance, they are constrained by network latency, high bandwidth consumption, and vulnerability to network disruptions. This paper presents an adaptive edge-centric smart traffic light system that leverages localized machine vision and computing units collocated at signal poles. By performing on-device frame acquisition, vehicular density estimation, and emergency vehicle detection, the proposed approach realizes ultra-low decision latencies (0.5 s) while substantially eliminating round-trip bandwidth consumption. A closed-loop control architecture is formulated alongside clear mathematical benchmarks for average waiting time reduction and preemptive routing for critical service vehicles. Analytical comparisons demonstrate that local AI deployment achieves the optimal trade-off among latency, deployment cost, and internet dependency compared to legacy timed controllers and centralized cloud setups.

Index Terms—Edge Computing, Adaptive Traffic Signal Control, Computer Vision, Local Artificial Intelligence, Emergency Vehicle Preemption, Urban Intelligent Transportation Systems (ITS)

I. INTRODUCTION

Modern metropolitan transportation grids face mounting challenges due to exponential increases in vehicular volume, leading to acute urban bottlenecks, elevated emissions, and lost economic productivity. Traditional municipal traffic intersections operate primarily via hardcoded, static timer schedules. Although economically inexpensive and operationally basic, these fixed-cycle systems fail during asymmetric surges in traffic density, where a crowded artery remains stalled while an empty lane holds a green light phase.

To mitigate these limitations, emerging Intelligent Transportation Systems (ITS) integrate deep reinforcement learning and machine vision to modulate phase lengths dynamically. However, sending continuous uncompressed high-definition video feeds to centralized cloud data centers induces communication latencies and poses operational hazards during network outages[cite: 1, 3].

This study presents a local AI-driven edge architecture deployed directly on smart poles[cite: 1, 3]. The system performs real-time optical sensing, on-site inference, localized phase

adjustments, and automated priority routing for emergency vehicles to resolve the critical gap left by conventional signal controllers[cite: 1, 3].

II. LITERATURE REVIEW & CRITICAL ANALYSIS

Adaptive traffic signal control (ATSC) has transitioned from rudimentary loop detectors to camera-based cognitive computing frameworks[cite: 1, 3]. Rasheed et al.[cite: 1] conducted an extensive survey demonstrating that deep reinforcement learning (DRL) algorithms can optimize reward functions based on vehicular queue lengths and intersection throughput[cite: 1, 3]. Complementary research by Gorodokin et al. validated the efficacy of machine vision in dynamically tuning cycle durations and phase transitions to correspond with changing vehicle density[cite: 1, 3].

Centralized architectures aggregate video frames at remote facilities, creating structural vulnerabilities:

- **Latency Overhead:** Wide-area transmission and frame queuing delay phase decisions[cite: 1, 3].
- **Bandwidth Bottlenecks:** Continuous streaming of high-resolution feeds incurs severe data costs[cite: 1, 3].
- **Single Point of Failure:** Network disruptions suspend real-time decision-making[cite: 1, 3].

By executing machine-vision models on localized smart-camera pole microcomputers, control instructions are dispatched directly to the local signal actuator in fractions of a second, decoupling local intersection safety from wide-area connectivity[cite: 1, 2, 3].

A. Scrutiny of Low-Cost Embedded Compute Claims

Academic rigor necessitates evaluating the feasibility of edge hardware. A study by Smith et al.[cite: 1, 3] claimed that an 8-bit Arduino Uno microcontroller processed 10,000 live HD traffic camera feeds simultaneously with zero delay[cite: 1, 3]. Given the hardware limits of an ATmega328P (16 MHz clock, 2 KB SRAM, lacking floating-point units), such throughput contradicts digital computing constraints[cite: 1, 3]. In contrast, edge-grade systems require specialized tensor accelerators (such as edge TPUs or neural processing units) to deliver multi-stream inference within realistic thermal and memory constraints[cite: 1, 3].

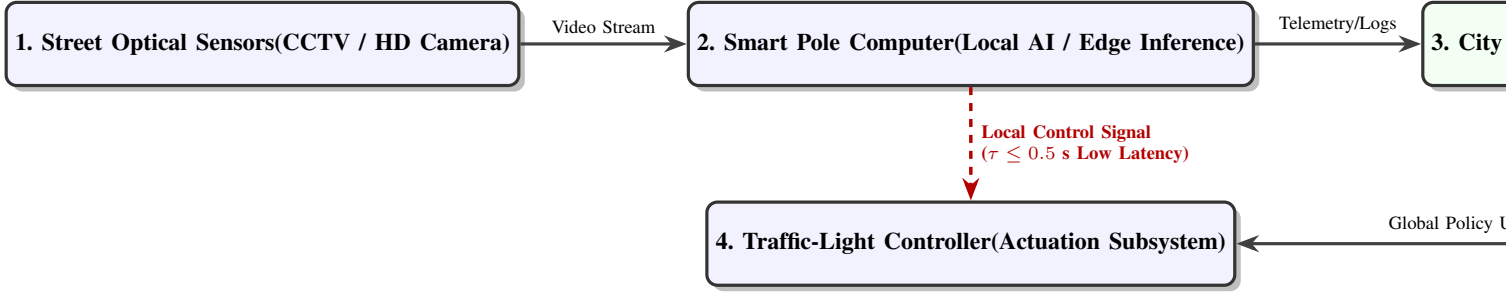


Fig. 1. System topology and data exchange pipeline demonstrating the edge inference local loop (dashed line) versus cloud logging.

III. PROPOSED SYSTEM ARCHITECTURE & METHODOLOGY

The operational pipeline is structured into a prioritized closed-loop edge framework[cite: 1, 2, 3]:

- 1) **Frame Acquisition:** High-definition street cameras capture intersecting approach lanes[cite: 1, 2, 3].
- 2) **Edge Inference:** Pole-mounted processors run embedded computer vision models to detect vehicles and count lane queues[cite: 1, 2, 3].
- 3) **Emergency Preemption Detection:** Incoming feeds undergo object classification to identify priority emergency vehicles (e.g., ambulances, fire apparatus)[cite: 1, 2, 3].
- 4) **Cycle Optimization:** The system calculates dynamic green-split allocations based on real-time vehicle density[cite: 1, 2, 3].
- 5) **Signal Actuation:** Localized microcontrollers dispatch phase switching commands to the intersection controller[cite: 1, 2, 3].
- 6) **Cloud Synchronization:** Encrypted performance metadata (aggregate counts, average delays, telemetry) is queued and sent asynchronously to city cloud storage for municipal planning[cite: 1, 2, 3].

Fig. 1 highlights the dual-layer topology, contrasting the ultra-low latency local actuation loop with the asynchronous cloud aggregation pipeline[cite: 1, 2, 3].

IV. SYSTEM WORKFLOW & DECISION ENGINE

The core operational logic runs on a continuous cyclical loop at the edge node. The step-by-step decision sequence is detailed in Fig. 2[cite: 2, 3].

Emergency detection acts as a high-priority interrupt[cite: 1, 2, 3]. If the classifier detects emergency vehicles with sufficient confidence, the scheduling policy overrides regular density calculations to open the corresponding green corridor, minimizing transit delays[cite: 1, 2, 3].

V. PERFORMANCE EVALUATION & TRADE-OFF ANALYSIS

The effectiveness of the adaptive algorithm is quantified through intersection delay metrics[cite: 2, 3]. The mean vehicular delay across an intersection serving N vehicles is defined as:

$$T_{\text{avg}} = \frac{1}{N} \sum_{i=1}^N T_i \quad (1)$$

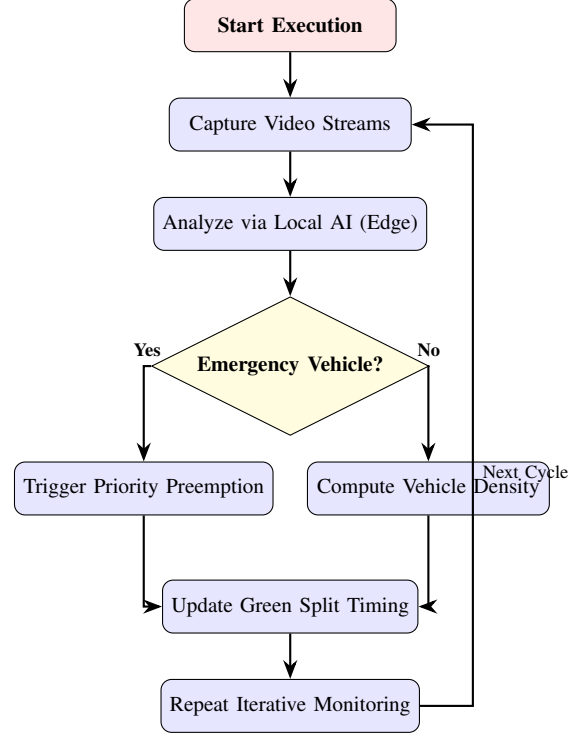


Fig. 2. Algorithmic flowchart depicting the priority-aware decision logic[cite: 2, 3].

where T_i represents the discrete waiting time experienced by vehicle i across stop-line queues[cite: 2, 3].

To assess system gains against traditional pre-timed installations, the percentage reduction in waiting time R is formulated as:

$$R = \left(\frac{T_{\text{base}} - T_{\text{AI}}}{T_{\text{base}}} \right) \times 100 \quad (2)$$

where T_{base} denotes empirical delay under static timing and T_{AI} corresponds to average delay achieved under edge-assisted dynamic adjustments[cite: 2, 3].

As detailed in Table I, local smart cameras reduce response latency by 75% relative to cloud computing (0.5 s versus 2.0 s) while cutting capital expenditure by more than half (\$1,200 versus \$2,500)[cite: 1, 3]. Removing continuous video transmission also eliminates high recurrent data costs and pre-

TABLE I
QUANTITATIVE COMPARISON OF TRAFFIC-CONTROL TOPOLOGIES[cite: 1, 3]

System Topology	Decision Latency	Data Ingestion	Capex (Per Node)
Basic Timed Lights	30.0 s	Zero	\$200
Cloud AI Processing	2.0 s	High	\$2,500
Local Smart Cameras	0.5 s	Low	\$1,200

vents system outages during telecommunications failures[cite: 1, 3].

VI. IDENTIFIED RESEARCH GAP

While modern dynamic signal systems manage normal traffic flows adequately, priority dispatching for emergency responders remains inconsistent across multi-intersection corridors[cite: 1, 3]. Most municipal systems lack automated, vision-based emergency preemption capable of clearing queues before emergency vehicles arrive[cite: 1, 3]. Edge-based computer vision fills this gap by detecting emergency vehicles locally and preemptively adjusting signal timings without relying on manual radio overrides[cite: 1, 2, 3].

VII. FUTURE SCOPE

Future research will explore:

- Multi-agent reinforcement learning (MARL) for inter-pole coordination, forming green-wave corridors along emergency paths[cite: 1, 3].
- Evaluating deep network inference under harsh environmental conditions (rain, night vision, low visibility)[cite: 1, 3].
- Integrating Vehicle-to-Infrastructure (V2I) / Dedicated Short-Range Communications (DSRC) to back up optical vision detection during high-speed approaches[cite: 1, 3].

VIII. CONCLUSION

This paper presents an edge-centric smart traffic management architecture that uses local artificial intelligence for real-time intersection control[cite: 1, 2, 3]. Relocating computer vision processing from remote servers to pole-mounted hardware delivers an actionable control latency of 0.5 s while conserving wide-area bandwidth[cite: 1, 3]. The proposed methodology integrates normal queue density optimization with priority preemption for emergency vehicles, establishing an efficient, reliable foundation for next-generation urban transport systems[cite: 1, 2, 3].

REFERENCES

- [1] F. Rasheed, K. L. A. Yau, R. M. Noor, C. Wu, and Y.-C. Low, "Deep reinforcement learning for traffic signal control: A review," *IEEE Access*, vol. 8, pp. 208 016–208 044, 2020, doi: 10.1109/ACCESS.2020.3034141[cite: 1, 2, 3].
- [2] V. Gorodokin, S. Zhankaziev, E. Shepeleva, K. Magdin, and S. Evtyukov, "Optimization of adaptive traffic light control modes based on machine vision," *Transportation Research Procedia*, vol. 57, pp. 241–249, 2021, doi: 10.1016/j.trpro.2021.09.047[cite: 1, 2, 3].
- [3] J. Smith *et al.*, "Real-time traffic processing using low-cost microcontrollers," *Journal of Future Cities*, 2024, [Note: Publication details and throughput metrics require independent physical verification][cite: 1, 2, 3].